

CHAPTER-1

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INTRODUCTION

1.1 Automated Waste Segregation

An Automated Waste Segregation System is a technological solution designed to sort and classify waste materials automatically. This system employs various technologies such as sensors, artificial intelligence, robotics, and mechanical sorting mechanisms to differentiate between different types of waste, including recyclables, organic waste, and non-recyclables.

The primary goal of these systems is to improve the efficiency and accuracy of waste management processes, reduce human intervention, and enhance recycling rates by ensuring that each type of waste is correctly processed and disposed of.

1.2 EXISTING SYSTEM

Some common methods include color-coded bins for different types of waste (e.g., blue for recyclables, green for organic waste, black for non-recyclable waste), community recycling centers, and educational campaigns to promote proper waste sorting. These systems aim to make it easier for individuals and communities to segregate their waste effectively, leading to more efficient recycling and waste management practices.

In basic manual sorting, workers manually separate waste into

different categories, such as metals, plastics, paper, and organic materials. This method relies solely on human effort and judgment without any mechanical aids.

1.3 PROBLEM IDENTIFICATION

- **High Operational Costs:** Manual segregation requires a significant number of workers, leading to high labor costs.
- **Physical Strain:** Workers are subject to physical strain and fatigue, which can reduce efficiency over time.
- **Human Error:** Sorting accuracy can be inconsistent due to human error, leading to incorrect segregation.
- **Exposure to Hazards:** Workers are exposed to hazardous materials, which can lead to health issues such as respiratory problems, skin infections, and injuries.
- **Bottlenecks:** Manual processes can create bottlenecks, especially when dealing with large volumes of waste.
- **Improper Segregation:** Inaccurate sorting can lead to improper disposal of recyclable materials, reducing recycling rates and increasing landfill usage.
- **Contamination:** Incorrectly segregated waste can contaminate recycling streams, rendering entire batches unusable.

1.4 PROPOSED SYSTEM

The proposed Waste Segregation Machine is an advanced automated system designed to efficiently and accurately segregate different types of waste, such as metal, plastic, paper, and organic materials. This system aims to improve the efficiency of waste management processes, reduce human labor, and increase recycling rates.

The proposed system offers several advantages over segregation process, including:

Higher Recycling Rates: Accurate segregation leads to higher recycling rates, reducing the amount of waste sent to landfills and promoting environmental sustainability.

Reduced Exposure to Hazards: Automation reduces the need for human workers to handle hazardous materials, minimizing the risk of injuries and health issues.

Continuous Operation: Automated systems can operate 24/7 without the need for breaks, leading to higher overall productivity.

Positive Public Image: Demonstrating a commitment to efficient and environmentally responsible waste management enhances the public image of waste management facilities and municipalities.

Reduced Exposure to Hazards: Automated systems minimize human contact with hazardous materials, reducing the risk of injuries and health issues for workers.

The proposed system has significant potential to improve the quality of life for workers in segregation process and technology towards the environment.

1.5 SUMMARY

An Automated Waste Segregation System utilizes sensors, AI, robotics, and mechanical sorting to efficiently classify waste into recyclables, organics, and non-recyclables, improving waste management accuracy and reducing human intervention. It addresses issues of high operational costs, physical strain, human error, hazard exposure, and bottlenecks found in manual sorting. The system promises higher recycling rates, reduced exposure to hazards, continuous operation, and an enhanced public image for waste management facilities. Ultimately, it aims to improve worker safety and promote environmental sustainability through more efficient waste processing.

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LITERATURE SURVEY

2.1 INTRODUCTION

A literature survey on waste segregation machines involves reviewing a variety of sources to understand the current state of a technology, methodologies, and applications in this field.

2.2 IOT BASED AUTOMATIC WASTE SEGREGATOR

“Iot based Automatic waste segregator” S.Lopes, et.al.,[5] In the year 2019 at a International Conference on Advances in Computing, Communication and control (ICAC3) at Maharashtra, India. He proposed that the framework automatically detects and segregates household waste, generated on a routine basis into dry, wet and metallic waste. Block diagram of this work shown below in fig.2.1

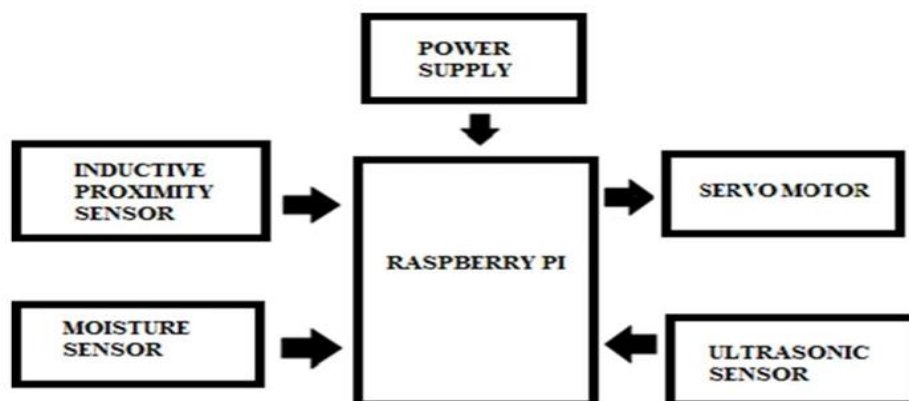


Figure 2.1 Paper 1 Block diagram

RESULT

The IoT-based automatic waste segregator automates the sorting of waste into different categories such as metal, plastic, and paper, significantly reducing the need for manual labor. Using sensors to identify waste types and a microcontroller to process this data, the system accurately directs waste to the appropriate bins via actuators. enables real-time monitoring and data logging through a network interface, providing valuable insights for optimizing waste management. The result is increased efficiency, improved sorting accuracy, and enhanced recycling rates, contributing to cost savings and environmental sustainability.

2.3 IMPLEMENTATION OF AN AUTOMATIC WASTE SEGREGATOR SYSTEM USING ARDUINO UNO

“Implementation of an Automatic Waste Segregator System using Arduino UNO”, D.Sunehra, et.al.,[10]. In the year of 2021 at 6th International Conference for convergence in Technology (12CT) at Maharashtra, India. He proposed that this type of system can be fabricated in larger dimension for using various places like education Institutions Officers Hotel. Block diagram of this work shown below in fig.2.2

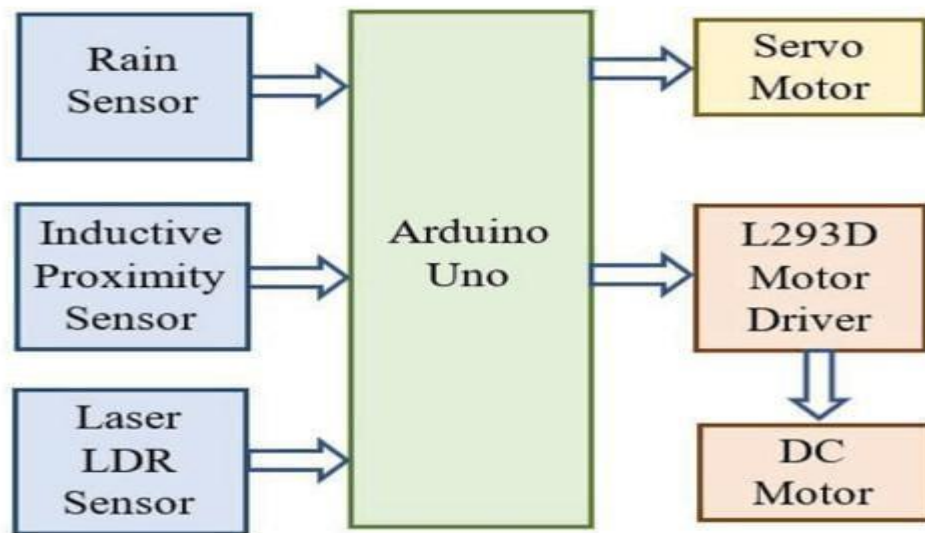


Figure 2.2 Paper 2 Block diagram

RESULT

An automatic waste segregator system using Arduino Uno starts with a power supply that provides necessary power to all components. The Arduino Uno acts as the central controller, processing inputs from various sensors and controlling outputs. An ultrasonic sensor detects the presence of waste, initiating the sorting process. The system uses an inductive sensor to identify metallic waste, a capacitive sensor to detect non-metallic waste like plastic, and a moisture sensor to detect organic waste based on moisture content. A motor driver interfaces the Arduino with motors, allowing the Arduino to control their operation. DC motors and servos move the conveyor belt and sorting mechanism. The conveyor belt transports waste to different sorting areas. Finally, segregated waste is directed into separate output bins for metal, plastic, and organic waste.

2.4 DEVELOPMENT OF AUTOMATIC WASTE SEGREGATOR WITH MONITORING SYSTEM

“Development of Automatic Waste Segregator with Monitoring System”, N.H.Kamarudin, et.al.,[4] In the year of 2019 at the 4th International Conference on Information Technology, Information Systems and Electrical Engineering (ICITISEE) at Yogyakarta, Indonesia. He proposed that this project is equipped with notifications systems to allot the janitor to ease the process of waste collection by receiving SMS. Block diagram of this work shown below in fig.2.3

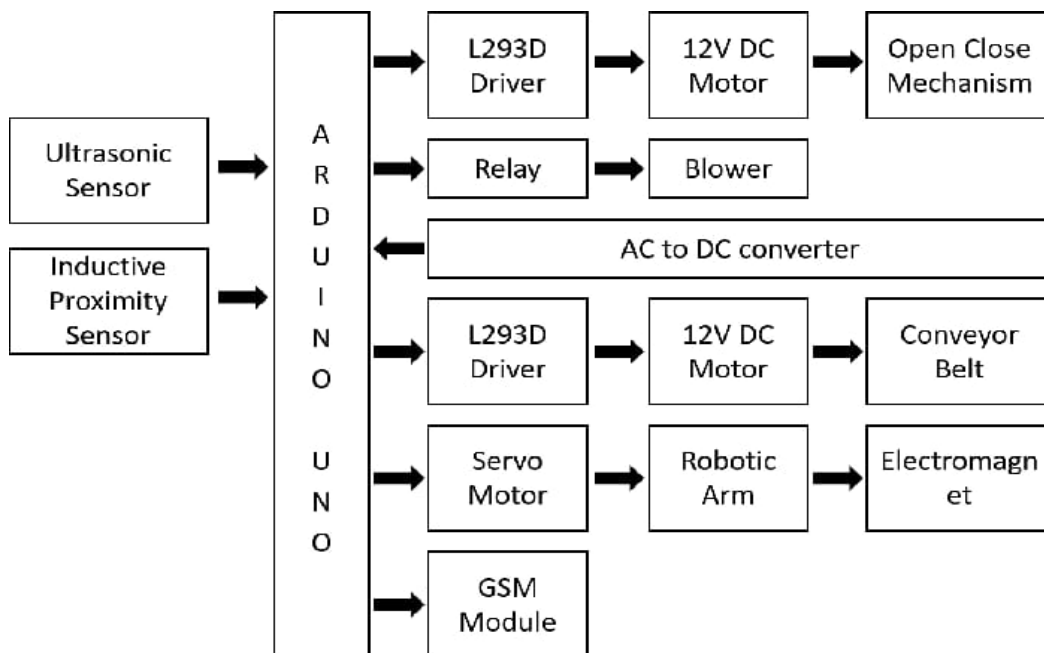


Figure 2.3 Paper 3 Block diagram

RESULT

The development of an Automatic Waste Segregator with Monitoring System not only addresses immediate waste management challenges but also contributes to sustainable development goals by promoting efficient resource use and environmental stewardship. It represents a significant advancement towards smarter, more efficient urban waste management solutions.

2.5 AUTOMATIC WASTE SEGREGATOR

“Automatic Waste Segrator”, A.Sharanya, et.al.,[8] In the year of 2017 at a International Conference on Advances in Computing, communications and Informatics (ICACCI) at Udupi, India. She proposed that this can be largely implemented in various municipal corporations and places like airports or railway station taking into consideration various factors such as reduction in man power and increase speed of Waste management. Block diagram of this work shown below in fig.2.4



Figure 2.4 Paper 4 Block diagram

RESULT

Implementing this type of system improves waste segregation practices, promotes recycling efforts, and contributes to environmental sustainability by ensuring that different types of waste are handled appropriately. The actual performance and effectiveness can vary based on the specific sensors, actuators, and algorithms used, as well as the quality of implementation and calibration.

2.6 SUMMARY

PAPER 1 – “Iot based Automatic waste segregator”
S.Lopes, et.al., at ICAC3 in 2019. The System uses sensors and microcontroller (e.g., Arduino, Raspberry Pi) to automatically detect and segregate household waste.

PAPER 2 – “Implementation of an Automatic Waste Segregator System using Arduino Uno”, D.Sunehra, et.al., at 12CT in 2021. The System uses Arduino UNO, Ultrasonic sensor, Servo Motor to segregate the output bins for metal, plastic and organic wastes.

PAPER 3 – “Development of Automatic Waste Segregator with Monitoring System”, N.H.Kamarudin, et.al., at ICITISEE in 2019. The System uses sensors(Ultrasonic and proximity sensor), Arduino, Servo Motor and GSM module. They proposed

this project is equipped with notification system in advancement of waste segregation.

PAPER 4 – “Automatic Waste Segrator”, A.Sharanya, et.al., at ICACCI in 2017. The system uses Sensors, Arduino and Actuators. They proposed this project in various municipal corporations for the reduction in man power and increase speed of Waste management.

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METHODOLOGY

3.1 INTRODUCTION

In this project, we propose the development of a Waste segregation machine aims to automate the sorting of waste materials into distinct categories such as organic, recyclable, and non-recyclable waste and this system also used for the sanitary workers to easily segregate the wastes from residential areas. This device is made up of Arduino UNO, Soil Sensor, The system leverages an Arduino UNO microcontroller, along with a suite of sensors and actuators, to accurately identify and sort various types of waste. The methodology encompasses the design, integration, and programming of the system components to achieve effective waste segregation.

3.2 PROPOSED BLOCK DIAGRAM

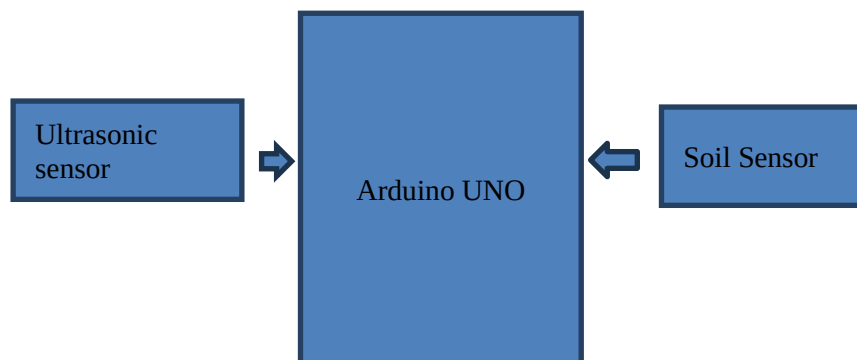


Figure 3.1 Proposed Block diagram

Figure 3.1 represents the system will use an ultrasonic sensor detects incoming waste and measures its distance, the soil sensor measures the moisture content of the waste and the Arduino processes the sensor data to classify the waste as wet or dry materials. This setup allows for an automated system that sorts waste into different categories based on distance and moisture content, improving efficiency and accuracy in waste management.

3.3 CIRCUIT DIAGRAM

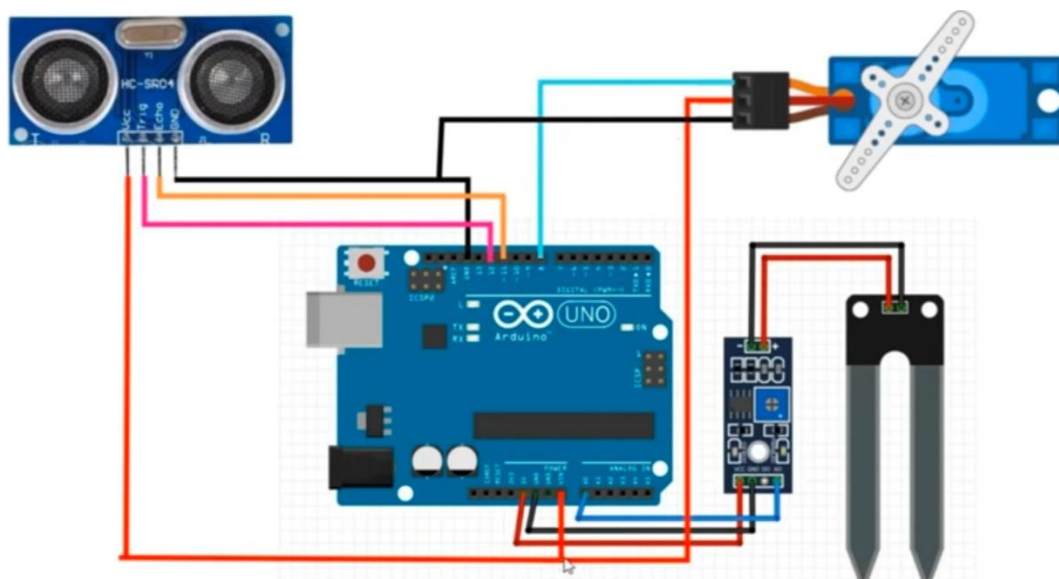


Figure: 3.2 Circuit Diagram

Figure 3.3 Circuit diagram involves an Arduino Uno, an ultrasonic sensor, a servo motor, and a soil moisture sensor. Here's a brief explanation of the connections

Ultrasonic Sensor (HC-SR04):

VCC: Connect to the 5V pin on the Arduino.

Trig: Connect to a digital I/O pin (for example, D9).

Echo: Connect to another digital I/O pin (for example, D10).

GND: Connect to the ground (GND) pin on the Arduino.

Servo Motor:

Power (Red wire): Connect to the 5V pin on the Arduino.

Ground (Black/Brown wire): Connect to the ground (GND) pin on the Arduino.

Signal (Yellow/White wire): Connect to a PWM-capable digital I/O pin (for example, D11).

Soil Moisture Sensor:

VCC: Connect to the 5V pin on the Arduino.

GND: Connect to the ground (GND) pin on the Arduino.

Analog Output (AO): Connect to an analog input pin (for example, A0).

Detailed Connections:

Power Supply: Ensure the 5V and GND connections from the Arduino are properly distributing power to the ultrasonic sensor, the servo motor, and the soil moisture sensor.

Signal Connections:

For the ultrasonic sensor, the Trig and Echo pins are used to measure distance by sending and receiving sound waves. The servo motor's signal pin receives PWM signals to control its position. The soil moisture sensor's analog output provides a voltage level proportional to the moisture level in the soil, which is read by the Arduino's analog input pin.

3.4 SUMMARY

This project involves the development of an automated waste segregation machine designed to sort waste materials into categories such as organic, recyclable, and non-recyclable. This system aims to assist sanitary workers in residential areas by simplifying and enhancing the efficiency of waste segregation.

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COMPONENTS

4.1 HARDWARE COMPONENTS

4.1.1 ARDUINO UNO

The Arduino Uno is a popularly used open-source micro-controller board that runs on ATmega 328P micro-controller. Arduino uno this board is developed by Arduino.cc which is an Italy based hardware company. This board contains a set of digital and analog I/ O data pins that are used to interface this board with other electronic components. Arduino Uno consists of 14 digital pins and 6 analog pins. This board can be programmed with the help of Arduino IDE (Integrated Development Environment) that supports embedded C, its back-end is constructed using JAVA. Uno consists of an USB port through which the code can be uploaded on to the board. This port can also be used to power the board by connecting it to a laptop, PC, etc. Along with a USB port, it also has a DC input power jack. An external battery of 9V can also be used to power Arduino board. The Arduino UNO is shown below in fig. 4.1.



Figure 4.1 Arduino UNO

KEY FEATURES:

Microcontroller: ATmega328P

Operating Voltage: 5V

Input Voltage (recommended): 7-12V

Input Voltage (limits): 6-20V

Digital I/O Pins: 14 (of which 6 provide PWM output)

Analog Input Pins: 6

DC Current per I/O Pin: 20 Ma

DC Current for 3.3V Pin: 50 mA

Flash Memory: 32 KB (ATmega328P) of which 0.5 KB used by bootloader

SRAM: 2 KB (ATmega328P)

EEPROM: 1 KB (ATmega328P)

Clock Speed: 16 MHz

Physical Characteristics**Dimensions:** 68.6 mm x 53.4 mm

Weight: Approximately 25 g

Power Supply: Can be powered via the USB connection or with an external power supply. The power source is selected automatically.

4. Connectivity

USB Connection: Used for power, programming, and communication.

Power Jack: For external power supply (7-12V).

ICSP Header: For programming the microcontroller directly using an external programmer.

Reset Button: Resets the microcontroller.

5. Pins and Interfaces

Digital I/O Pins: 14 digital pins (pins 0-13) that can be used as input or output using the pinMode(), digitalWrite(), and digitalRead() functions.

PWM: Pins 3, 5, 6, 9, 10, and 11 provide 8-bit PWM output with the analog write() function.

Analog Input Pins: 6 analog inputs (pins A0-A5), each of which provides 10 bits of resolution (i.e., 1024 different values).

Power Pins:

Vin: The input voltage to the Arduino board when it's using an external power source (as opposed to 5V from the USB connection or other regulated power source).

5V: The regulated power supply used to power the microcontroller and other components on the board.

3.3V: A 3.3V supply generated by the onboard regulator. Maximum

current draw is 50 mA.

GND: Ground pins.

IOREF: This pin on the Arduino board provides the voltage reference with which the microcontroller operates

4.1.2 SOIL SENSOR

- Soil moisture sensor measures the water content in the soil. In the provided circuit diagram, the sensor appears to be a capacitive or resistive type sensor. Here's a detailed explanation of how the soil moisture sensor works and how it is connected in the circuit. The Soil Sensor is shown below in fig. 4.2.

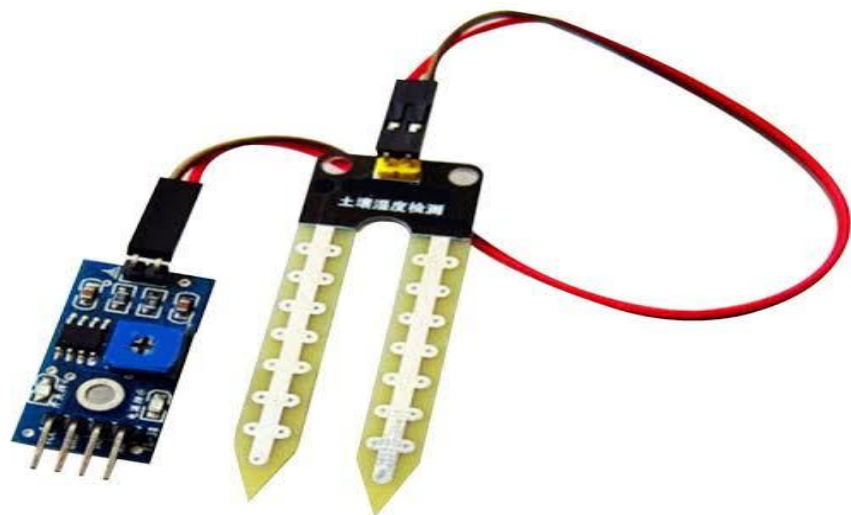


Figure 4.2 Soil Sensor

Components of the Soil Moisture Sensor:

Probe:

Capacitive Type: Uses a capacitive method to detect soil moisture, which is less prone to corrosion compared to resistive types.

Resistive Type: Consists of two exposed electrodes that measure the resistance between them. Higher water content decreases the resistance.

Control Board:

Voltage Regulator: Ensures a stable voltage supply to the sensor.

Analog Output (AO): Provides an analog voltage output proportional to the moisture level in the soil.

Digital Output (DO): Often used for a simple threshold-based output, not shown in the given diagram.

Connection in the Circuit:

VCC Pin: Connects to the 5V output on the Arduino. Powers the sensor.

GND Pin: Connects to one of the GND pins on the Arduino. Completes the circuit by providing a reference ground.

Analog Output (AO) Pin: Connects to an analog input pin on the Arduino (e.g., A0). Sends a variable voltage signal to the Arduino, representing the soil moisture level.

4.1.3 SERVO MOTOR

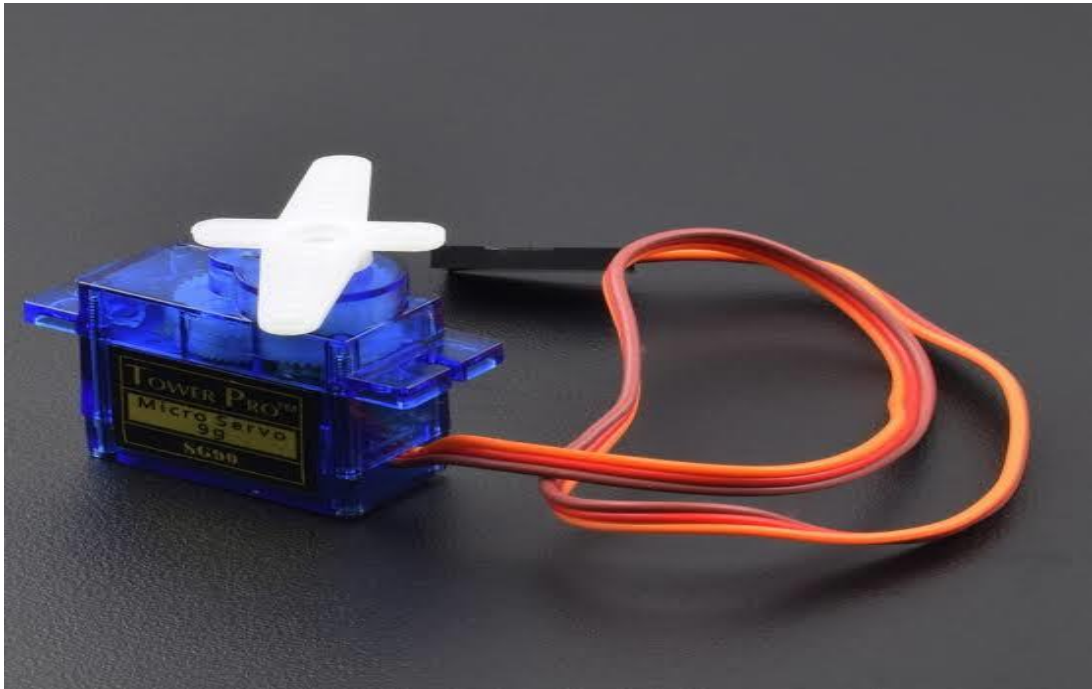


Figure 4.3 Servo Motor

Servo motor is a rotary or linear actuator that allows for precise control of angular or linear position, velocity, and acceleration in a mechanical system. It constitutes part of a servomechanism, and consists of a suitable motor coupled to a sensor for position feedback and a controller.

The servo motor is to convert the control signal of the controller into the rotational angular displacement or angular velocity of the motor output shaft. Servo motor is used to drive the joints. The Servo Motor is shown in fig 4.3.

4.1.4 ULTRASONIC SENSOR



Figure 4.4 Ultrasonic sensor

Ultrasonic Sensor is a device that determine the level of waste. The HC-SR04 is a popular ultrasonic sensor used for measuring distances. It uses sound waves to determine the distance to an object, making it useful for a variety of applications, such as obstacle detection, distance measurement, and automation projects. The Ultrasonic Sensor is shown above in fig 4.4.

Components of Ultrasonic Sensor

Transmitter (Trigger): Emits ultrasonic sound waves.

Receiver (Echo): Detects the reflected sound waves.

Control Circuit: Manages the timing and processing of the signals.

The sensor emits an ultrasonic pulse from the transmitter, which travels through the air. When it encounters an object, the pulse is reflected back and detected by the receiver. The time it takes for the echo to return is measured and used to calculate the distance to the object.

Pin Configuration

VCC: Connects to the 5V pin on the Arduino.

Trig: Trigger pin, used to send the ultrasonic pulse.

Echo: Echo pin, used to receive the reflected pulse.

GND: Connects to the ground pin on the Arduino.

By connecting the HC-SR04 ultrasonic sensor to the Arduino as shown in the circuit diagram, you can measure distances and use this data to control other components like a servo motor or trigger specific actions.

4.2 SOFTWARE COMPONENTS

4.2.1 ARDUINO IDE

Arduino IDE, IDE stands for “Integrated Development Environment”: it is an official software introduced by Arduino.cc, that is mainly used for editing, compiling and uploading the code in the Arduino Device. Almost all Arduino modules are compatible with this software that is an open source and is readily available to install and start compiling

the code on the go. In this article, we will introduce the Software, how we can install it, and make it ready for developing applications using Arduino modules.

1. Arduino IDE is an open-source software that is mainly used for writing and compiling the code into the Arduino Module.
2. It is an official Arduino software, making code compilation too easy that even a common person with no prior technical knowledge can get their feet wet with the learning process.
3. It is easily available for operating systems like MAC, Windows, Linux and runs on the Java Platform that comes with inbuilt functions and commands that play a vital role for debugging, editing and compiling the code in the environment.
4. A range of Arduino modules available including Arduino Uno, Arduino Mega, Arduino Leonardo, Arduino Micro and many more.
5. Each of them contains a microcontroller on the board that is actually programmed and accepts the information in the form of code.
6. The main code, also known as a sketch, created on the IDE platform will ultimately generate a Hex File which is then transferred and uploaded in the controller on the board.

7. The IDE environment mainly contains two basic parts: Editor and Compiler where former is used for writing the required code and later is used for compiling and uploading the code into the given Arduino Module.
8. This environment supports both C and C++ languages.

ARDUINO IDE

Software can be downloaded from Arduino main website. As I said earlier, the software is available for common operating systems like Linux, Windows, and MAC OS, we select to download the correct software version that is easily compatible with our operating system. The Arduino IDE figure shown in below fig.4.5.

Details on IDE:

The IDE environment is mainly distributed into three sections

1. Menu Bar
2. Text Editor
3. Output Pane

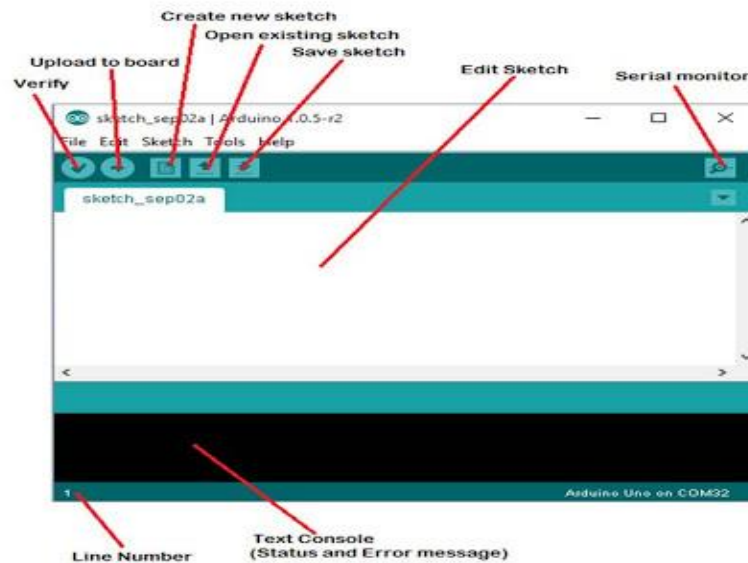


Figure: 4.5 Arduino IDE

The bar appearing on the top is called Menu Bar that comes with five different options as follow:

- File – You can open new window for writing the code or open an existing one. Following table shows the number of further subdivisions the file option is categorized into.
 - Menu Bar
 - Text Editor
 - Output Pane
- Edit – Used for copying and pasting the code with further modification for font
- Sketch – For compiling and programming
- Tools – Mainly used for testing projects. The Programmer section in this panel is used for burning a bootloader to the new microcontroller.

- Help – In case you are feeling skeptical about software, complete help is available from getting started to troubleshooting. The Six Buttons appearing under the Menu tab are connected with the running program as follow.
- The Six Buttons appearing under the Menu tab are connected with the running program as follow. A separate pop-up window appears as an independent terminal and plays a vital role for sending and receiving the Serial Data. You can also go to the Tools panel and select Serial Monitor pressing Ctrl+Shift+M all at once will open the Serial Monitor.
- The Serial Monitor will actually help to debug the written Sketches where you can get a hold of how your program is operating. Your Arduino Module should be connected to your computer by USB cable in order to activate the Serial Monitor.
- You need to select the baud rate of the Arduino Board you are using right now.

Program Structure:

- Declarations
- Variables

Whenever you're using Arduino, you need to declare global variables and instances to be used later on. In a nutshell, a variable allows you to name and store a value to be used in the future. For example, you would store data acquired from a sensor in order to use it later. To declare a variable, you simply define its type, name and initial value. It's worth mentioning that declaring global variables isn't an absolute necessity. However, it's advisable that you declare your variables to make it easy to utilize your values further down the line.

- Instances

In software programming, a class is a collection of functions and variables that are kept together in one place. Each class has a special function known as a constructor, which is used to create an instance of the class. In order to use the functions of the class, we need to declare an instance for it.

- Setup ()

Every Arduino sketch must have a setup function. This function defines the initial state of the Arduino upon boot and runs only once. Here we'll define the following:

1. Pin functionality using the pin Mode function
2. Initial state of pins
3. Initialize classes
4. Initialize variables

5. Code logic Loop () The loop function is also a must for every Arduino sketch and executes once setup () is complete. It is the main function and as its name hints, it runs in a loop over and over again. The loop describes the main logic of your circuit

Serial Plotter Arduino:

Serial plotter is another component of the Arduino IDE, which allows you to generate a real-time graph of your serial data. The serial plotter makes it much easier for you to analyze your data through a visual display. You're able to create graphs, negative value graphs, and conduct waveform analysis

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5.1 RESULTS AND DISCUSSION

Waste segregation covers the various aspects of the implemented system, its impact, challenges faced, and potential areas for improvement. It helps for the improvement of recycling rates, Waste diversion and Resource recovery. Here the environmental benefits of waste segregation, including reduced greenhouse gas emissions, conservation of natural resources, and prevention of pollution and habitat destruction.

Table: 5.1 Result of Waste Segregation Machine

S.NO	Waste Materials	Segregated Bins
1.	Plastic Items	Non-biodegradable bin
2.	Vegetable Peels	Biodegradable bin
3.	Flower	Biodegradable bin
4.	Glass piece	Non-biodegradable bin
5.	Rubber	Non-biodegradable bin

Table 5.1 represents the results of the waste segregation of some waste materials. It Segregates the biodegradable and non-biodegradable waste materials into separate bins. The plastic materials, glass pieces and rubber are categorized as non-biodegradable and the vegetable peels and flowers are categorized as biodegradable wastes. The Waste Segregation Machine is shown in below fig.5.1.

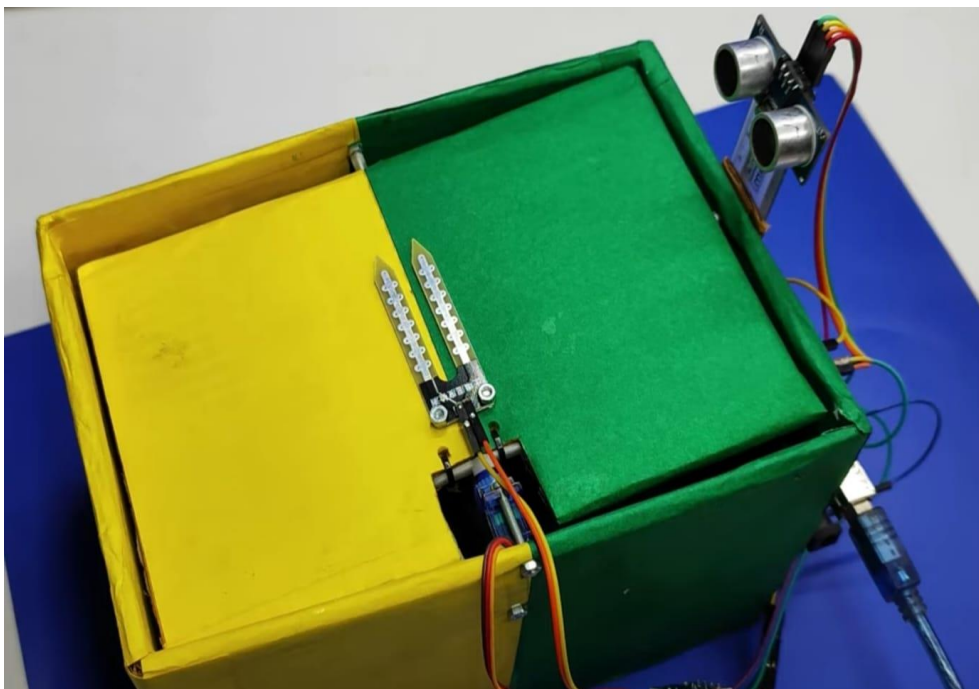


Figure 5.1 Waste Segregation Machine

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CONCLUSION

In conclusion, the Waste Segregator machine is a crucial step forward in addressing the global waste crisis. It aims to address the critical issue of efficient waste segregation. By automating the sorting process, this machine reduces the reliance on manual labor and increases the accuracy and speed of waste classification. This, in turn, facilitates better recycling practices and reduces the environmental footprint of waste disposal. Its adoption and widespread use can lead to a cleaner environment, more sustainable resource utilization, and a more resilient waste management infrastructure. As technology continues to evolve, further enhancements and innovations in waste segregation will undoubtedly emerge, driving us closer to a zero-waste future.

FUTURE WORK

Connecting the machine to the Internet of Things (IoT) for better monitoring and management. IoT sensors could provide data on fill levels, operational status, and performance metrics, enabling predictive maintenance and optimizing operational efficiency. Incorporating robotic arms or sorting mechanisms that can handle delicate or hazardous materials with precision. Robots can also assist in the sorting process, especially for items that are difficult for the machine to handle automatically.

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